

15/9/25.

LINEAR PROGRAMMING PROBLEM

It has certain target based situations.

- A model used for optimum allocation of scarce / limited resources to competing products / activities under such assumptions as certainty, linearity & fixed technology.

(a) R/L b/w variable & constraints - linear.

(b) model must have objective fn.
" structural constraints
" non negative constraints.

model any situation mathematically

Machines	Products		Availability in hours
	x	y	
M1	0	2	50
M2	1	2	25
M3	1	1	15
profit in Rs. per unit	5	4	

Here to maximize the profit, the objective function would be $5x + 4y$.

thus Maximize $Z = 5x + 4y$.

Constraints given $\downarrow$ profit. $\Rightarrow$ here the profit has to be maximum.

for machine 1 $0x + 2y \leq 50$
 machine 2 $1x + 2y \leq 25$
 machine 3 $1x + 1y \leq 15$
 Both x and y ≥ 0 .

LPP $\rightarrow$ Graphical solutions :-

Z Equation line method.

$$\begin{aligned} \text{Max. } Z &= 3x_1 + 5x_2 \\ x_1 + 2x_2 &\leq 2000 \\ x_1 + x_2 &\leq 1500 \\ x_2 &\leq 600 \\ x_1, x_2 &\geq 0. \end{aligned}$$

} P.T.O.

Qn.).

$$\text{Max. } Z = 3x_1 + 5x_2$$

subjected To,

$$x_1 + 2x_2 \leq 2000.$$

$$x_1 + x_2 \leq 1500$$

$$x_2 \leq 600$$

$$x_1, x_2 \geq 0$$

Soln:

$$Z = 3x_1 + 5x_2$$

$$x_1 + 2x_2 = 2000 \quad \text{--- (1)}$$

$$x_1 + x_2 = 1500 \quad \text{--- (2)}$$

$$x_2 = 600 \quad \text{--- (3)}$$

$$x_1 = 0 \Rightarrow x_2 = 1000.$$

$$(0, 1000)$$

$$x_2 = 0 \Rightarrow x_1 = 2000$$

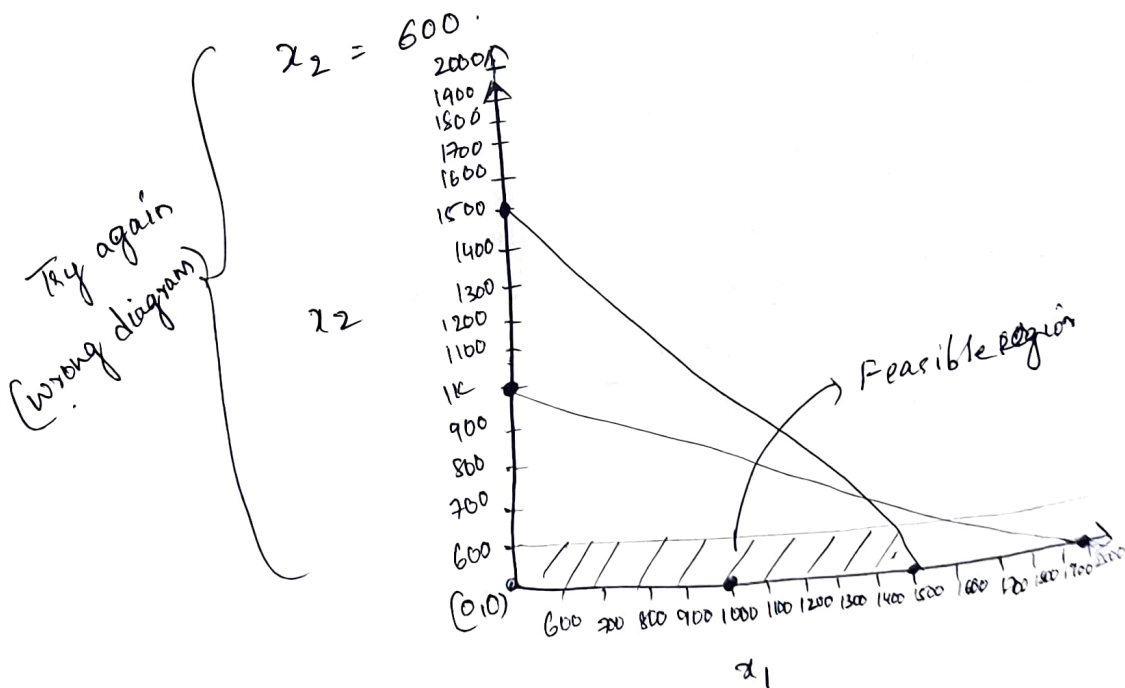
$$(2000, 0)$$

$$x_1 = 0 \Rightarrow$$

$$x_2 = 1500 \quad (0, 1500)$$

$$x_2 = 0 \Rightarrow$$

$$x_1 = 1500 \quad (1500, 0)$$



consider $z = 3x_1 + 5x_2$
 put $z = 0$
 $3x_1 + 5x_2 = 0 \quad \} \Rightarrow$

$$\frac{x_1}{x_2} = \frac{-5}{3}$$

$$\frac{x_1}{x_2} = \frac{-5}{3}$$

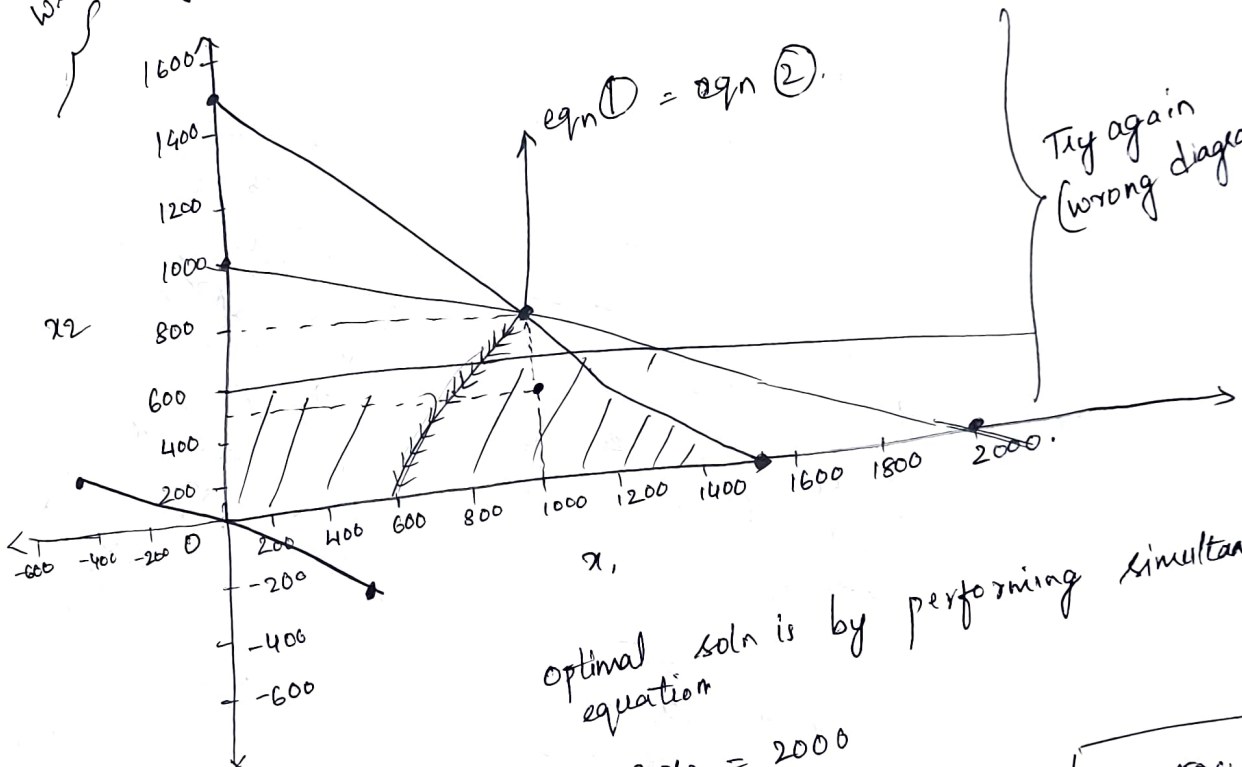
multiply by 100s.

$$\frac{x_1}{x_2} = \frac{-500}{300} = \frac{500}{-300}$$

points are $(-500, 300)$, $(500, -300)$

or $5x_2 = -3x_1$
 $\frac{5}{-3} = \frac{x_1}{x_2}$

wrong diagram
 Try again



optimal soln is by performing simultaneous equation

$$\begin{array}{r} x_1 + 2x_2 = 2000 \\ -x_1 + x_2 = 1500 \\ \hline x_2 = 500 \end{array}$$

thus $\begin{cases} x_2 = 500 \\ x_1 = 1000 \end{cases}$

feasible point
 $(1000, 500)$

$$\begin{array}{r} x_1 + x_2 = 1500 \\ x_2 + 500 = 1500 \\ x_2 = 1500 - 500 \\ x_2 = 1000 \end{array}$$

$$\begin{array}{r} x_1 + x_2 = 1500 \\ x_1 + 500 = 1500 \\ x_1 = 1500 - 500 \\ x_1 = 1000 \end{array}$$

thus the points are

$$(0,0) = 3(0) + 5(0) = 0.$$

$$(0,600) = 3(0) + 5(600) = 3000$$

$$(0,1500) = 3(0) + 5(1500) = 4500$$

$$(800,600) = 3(800) + 5(600) = 5400$$

$$(1000,500) = 3(1000) + 5(500) = 5500$$

Hence the max. value here is 5500, hence
the most feasible point is (1000, 500)

* MINIMIZATION PROBLEM

$$\text{Min } Z = 20x_1 + 10x_2$$

constraints:

$$x_1 + 2x_2 \leq 40.$$

$$3x_1 + x_2 \geq 30.$$

$$4x_1 + 3x_2 \geq 60.$$

$$x_1, x_2 \geq 0.$$

on solving (1) & (2)

$$\begin{array}{r} x_1 + 2x_2 = 40. \\ (-) 6x_1 + 2x_2 = 60. \\ \hline -5x_1 = -20 \end{array}$$

$$x_1 = 4$$

$$4 + 2x_2 = 40$$

$$2x_2 = 36$$

$$x_2 = 18$$

$$(4, 18)$$

$$\begin{array}{r} (1) \\ (2) \times 3 \\ \hline 6x_1 + 2x_2 = 30 \\ 4x_1 + 3x_2 = 60 \\ \hline -2x_1 - x_2 = -30 \\ \Rightarrow x_1 = 30 \end{array}$$

$$4 + 3$$

$$\begin{array}{r} 3x_1 + x_2 = 30 \\ 4x_1 + 3x_2 = 60 \\ \hline -x_1 - 2x_2 = -30 \\ 18x_1 + x_2 = 30 \\ \hline 17x_1 = 0 \\ x_1 = 0 \end{array}$$

$$x_1 + 2x_2 = 40$$

$$x_1 = 0$$

$$x_2 = 20$$

$$(0, 20)$$

$$x_1 + 0 = 40$$

$$(40, 0)$$

$$3x_1 + x_2 = 30$$

$$x_1 = 0$$

$$x_2 = 30$$

$$(0, 30)$$

$$3x_1 + 0 = 30$$

$$x_1 = 10$$

$$(10, 0)$$

$$4x_1 + 3x_2 = 60$$

$$3x_2 = 60$$

$$x_2 = 20$$

$$x_1 = 0$$

$$(0, 20)$$

$$x_2 = 0$$

$$4x_1 + 0 = 60$$

$$x_1 = 15$$

$$(15, 0)$$

$$20x_1 + 10x_2$$

points are

$$(0, 0) \Rightarrow 0 //$$

$$(0, 20) \Rightarrow 200 //$$

$$* (40, 0) \Rightarrow 800 //$$

$$* (15, 0) = 300 //$$

$$* (4, 18) = 260 //$$

$$(0, 30) = 300 //$$

$$(10, 0) = 200$$

$$* (6, 12) \Rightarrow 240 //$$

$$\begin{array}{r} 80 + 180 \\ = 180 \\ 80 \\ \hline 260 \end{array}$$

$$120 + 120 \Rightarrow 240$$

feasible point ~~(40, 0)~~ (6, 12)

110 (1) Min $Z = 0.3x_1 + 0.9x_2$ (2) points

$$x_1 + x_2 \geq 1400$$

$$0.21x_1 - 0.3x_2 \leq 0$$

$$0.03x_1 - 0.01x_2 \geq 0$$

$$x_1, x_2 \geq 0$$

$$x_1 + x_2 = 1400$$

(2) Max $Z = 40x_1 + 35x_2$ (4) points
constraints: (Redundant constraint)

$$2x_1 + 3x_2 \leq 60$$

$$4x_1 + 3x_2 \leq 96$$

$$3x_1 + 3x_2 \leq 105$$

$$x_1, x_2 \geq 0$$

* (3) Maximize optimal soln:

$$\text{Max: } Z = 8x_1 + 16x_2 \quad (5 \text{ points})$$

$$\text{constraints: } x_1 + x_2 \leq 200$$

$$x_2 \leq 125$$

$$3x_1 + 6x_2 \leq 900$$

$$x_1, x_2 \geq 0$$

* Infeasibility
↓

occurs when there is no feasible region.

Simplex method :-

* slack variable

Non negative variable is added to the left hand side of the constraint to convert $\leq$ to $=$ an equation is called a slack variable.

Given:-

$$3x_1 + x_2 \leq 2$$

$$\leq \Rightarrow +$$

$$x_1 + x_2 \geq 5$$

$$\geq \Rightarrow -$$

$$x_1, x_2 \geq 0.$$

* surplus variable (variable)

$$3x_1 + x_2 - s_1 = 2$$

$$x_1 + x_2 - s_2 = 5$$

$$x_1, x_2, s_1, s_2 \geq 0.$$

* Maximization & Minimization

1) Using simplex method, Maximize

$$Z = 3x_1 + 5x_2 + 4x_3.$$

subject to,

$$2x_1 + 3x_2 \leq 8.$$

$$2x_2 + 5x_3 \leq 10$$

$$3x_1 + 2x_2 + 4x_3 \leq 15.$$

convert to slack variable $x_1, x_2 \geq 0.$

$$2x_1 + 3x_2 + s_1 = 8$$

$$2x_2 + 5x_3 + s_2 = 10$$

$$3x_1 + 2x_2 + 4x_3 + s_3 = 15$$

$$Z = 3x_1 + 5x_2 + 4x_3 + 0s_1 + 0s_2 + 0s_3$$

$$\boxed{s_1, s_2, s_3 \geq 0}$$

14 15 16 17 18 19

$C_j \rightarrow$ coefficient of x_i and s_i in objective function

C_B	C_j Basic variable	x_1	x_2	x_3	s_1	s_2	s_3	RHS	Ratio
0	s_1	2	<u>3</u> (key element)	0	1	0	0	8	$8/3 = 2.66$
0	s_2	0	2	5	0	1	0	10	$10/2 = 5$
0	s_3	3	2	4	0	0	1	15	$15/2 = 7.5$
$Z_j = \sum C_B(a_{ij})$		0	0	0	0	0	0	0	
$C_j - Z_j$		3	<u>5</u> (max)	4	0	0	0		

$\downarrow$
max.

$$\begin{array}{r} 3 \overline{) 8} \\ \underline{6} \\ 2 \\ \underline{6} \\ 2 \\ \underline{2} \\ 0 \end{array}$$

Intersection of key row & key column is key element.

1st Iteration:

C_B	C_j Basic variable	x_1	x_2	x_3	s_1	s_2	s_3	RHS	Ratio
5	x_2	2	3	0	1	0	0	8/3	-
0	s_2	$\frac{0 - 2 \times 2}{3} = -4/3$	2	$5 - \frac{2 \times 0}{3} = 5$	$0 - \frac{2 \times 1}{3} = -2/3$	$1 - \frac{0 \times 2}{3} = 1$	$0 - \frac{0 \times 2}{3} = 0$	14/3	14/5 min
0	s_3	$3 - \frac{2 \times 2}{3} = 5/3$	2	$4 - \frac{2 \times 0}{3} = 4$	$0 - \frac{1 \times 2}{3} = -2/3$	$0 - \frac{0 \times 2}{3} = 0$	$1 - \frac{0 \times 2}{3} = 1$	29/3	29/12
$Z_j = \sum C_B(a_{ij})$		10/3	15/3	0	5/3	0	0	40/3	-
$C_j - Z_j$		-1/3	0	<u>4</u> (max)	-5/3	0	0		

S_i in objective function

6/10/25

Minimization Problem

→ looks for the maximum element (i.e. the

Minimize $z = x_1 - 3x_2 + 3x_3$
subjected to,

maximum row as well as the maximum column

$$3x_1 - x_2 + 2x_3 \leq 7$$

$$2x_1 + 4x_2 \geq -12$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10.$$

std lpp

Maximize,

$$z^* = -x_1 + 3x_2 - 3x_3 + 0s_1 + 0s_2 + 0s_3.$$

$$3x_1 - x_2 + 2x_3 + s_1 = 7 \quad \text{--- (1)}$$

$$2x_1 + 4x_2 - s_2 = -12 \quad \text{--- take -ve on both sides}$$

$$-2x_1 - 4x_2 + s_2 = 12 \quad \text{--- (2)}$$

$$-4x_1 + 3x_2 + 8x_3 + s_3 = 10 \quad \text{--- (3)}$$

C_j
Basic variable

0 s_1

0 s_2

0 s_3

Z_j

$C_j - Z_j$

	-1	3	-3	0	0	0	RHS
x_1	3	x_2	x_3	s_1	s_2	s_3	
	3	-1	2	1	0	0	7
	-2	-4	0	0	1	0	12
	-4	3	8	0	0	1	10
	0	0	0	0	0	0	0

3

max col.

Ratio

-

-

10/3
max row

des:

tio

0/3
x 70W

* Two phase - method

- when '=' add artificial variable.
- when '≥' subtract surplus variable and add artificial variable.
- when '≤' add slack variable

2 phases / steps :-

phase 1 :-

Minimize
 $a_1 + a_2 + \dots + a_n$

phase 2 :-

consider original LPP.

Qn) :- Max, $Z = -2x_1 - x_2$

subjected to :-

$$3x_1 + x_2 = 3$$

$$4x_1 + 3x_2 \geq 6$$

$$x_1 + 2x_2 \leq 4$$

$$x_1, x_2 \geq 0.$$

Converting the above equation to standard form.

Max, $Z = -2x_1 - x_2 + 0s_1 + 0s_2 + a_1 + a_2$

subjected to

$$3x_1 + x_2 + a_1 = 3$$

$$4x_1 + 3x_2 - s_1 + a_2 = 6$$

$$x_1 + 2x_2 + s_2 = 4$$

$s_1, s_2 \rightarrow$ slack variable

$a_1, a_2 \rightarrow$ artificial variables.

$$2 - \frac{1}{3} \Rightarrow \frac{6-1}{3} = \frac{5}{3}$$

phase 1 :- Minimize, Min $*2 = a_1 + a_2$

$$3x_1 + x_2 + a_1 = 3$$

$$4x_1 + 3x_2 - s_1 + a_2 = 6$$

$$x_1 + 2x_2 + s_2 = 4$$

find min value wrt. column, after which the min value wrt to the row is also to be considered.

$$1 - 0$$

$$0 - 1/3$$

$$4 - \frac{3}{3} = \frac{8}{3}$$

1st iteration :-

key value $\rightarrow 3$.

C_j		
	Basic variable	
C_B		
1	a_1	
1	a_2	
0	s_2	
	Z_j	
	$C_j - Z_j$	

x_1	x_2	s_1	s_2	a_1	a_2	RHS	Ratio
							min
3	1	0	0	1	0	3	①
4	3	-1	0	0	1	6	3/2
1	2	1	0	0	0	4	4
7	4	-1	0	1	1	9	
-7	-4	1	0	0	0		

$$1 + \frac{1}{3} \Rightarrow \frac{4}{3}$$

Key element is 3.

min value key value $\rightarrow 5/3$

2nd iteration

$$R_1 \rightarrow R_1/3$$

$$R_2 \rightarrow R_2 - 4R_1/3$$

$$R_3 \rightarrow R_3 - R_1/3$$

C_j		
	Basic variable	
C_B		
	x_1	
	a_2	
	s_2	
	Z_j	
	$C_j - Z_j$	

x_1	x_2	s_1	s_2	a_1	a_2	RHS	ratio
1/3	0	0	1/3	0	1	3	
5/3	-1	0	-4/3	1	2	6	6/5
5/3	0	0	-1/3	0	3	9/5	
-5/3	-1	0	-4/3	1	2		
-5/3	1	0	7/3	0			

min

$$\frac{3 - \frac{4 \times 1}{3}}{\frac{1}{3}} = \frac{3 - \frac{4}{3}}{\frac{1}{3}} = \frac{\frac{9-4}{3}}{\frac{1}{3}} = \frac{5}{1} = 5$$

$$R_2 \rightarrow \frac{3}{5} R_2$$

$$R_1 \rightarrow R_1 - \frac{R_2}{5}$$

	C_j	0	0	0	0	1	1	
	B.V.	x_1	x_2	s_1	s_2	a_1	a_2	RHS ratio
0	x_1	1	0	$1/5$	0	$9/5$	$-1/5$	$3/5$
0	x_2	0	1	$-3/5$	0	$-4/5$	$3/5$	$6/5$
0	s_2	0	$5/3 \times 3/5$	0	0	$-1/3 \times 4/5$	0	$3 \times 3/5$
0	s_2	0	0	1	1	1	-1	1
	Z_j	0	0	0	0	0	0	0
	$C_j - Z_j$	0	0	0	0	1	1	

As $C_j - Z_j \geq 0$, physical soln is being obtained.
 and $a_1 = 0, a_2 = 0$.
 feasible soln, thus columns a_1 & a_2
 can be removed.

phase II. Consider original LPP and remove the
 columns of a_1 and a_2 as $a_1 = 0$ and $a_2 = 0$.

Max $Z = -2x_1 - x_2$
 subjected to
 $3x_1 + x_2 = 3$.

$$\text{Max } Z = -2x_1 - x_2 + 0s_1 + 0s_2$$

<u>C_j</u>		-2	-1	0	0	
	<u>B.V.</u>	x_1	x_2	s_1	s_2	RHS
<u>C_B</u>	x_1	1	0	$1/5$	0	$3/5$
-2	x_2	0	1	$-3/5$	0	$6/5$
-1	s_2	0	0	1	1	1
0	s_1	0	0	$1/5$	0	$-12/5$
	Z_j	-2	-1	$1/5$	0	
	$C_j - Z_j$	0	0	$-1/5$	0	

As $C_j - Z_j \leq 0$, feasible soln is obtained.

$$x_1 \rightarrow 3/5$$

$$x_2 \rightarrow 6/5$$

$$Z_j \rightarrow \underline{\underline{-12/5}}$$

} feasible soln.

PROBABILITY

Random exp $\rightarrow$ when result cannot be determined with certainty before hand.

Sample space, $S \rightarrow$ non empty set that contains all possible outcomes of a random exp.

Event $\rightarrow$ subset of sample space.

Null event $\rightarrow$ when event has no elements in sample space.
 $\hookrightarrow$ case (2) when event has all elements in the sample space.

3 approaches for probability

classical approach $\rightarrow$ classical $\rightarrow$ (for cases $\div$ exhaustive cases)
 $\rightarrow$ Statistical
 $\rightarrow$ Approximative.

$$P(A') + P(A) = 1.$$

$$P(A') = \frac{n-m}{n}$$

$$P(A) = \frac{m}{n}.$$

conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (P(B) \neq 0)$$

Q1).

$\{1, 2, 3, 4, 5, 6\}$

odd $\{1, 3, 5\}$

$\rightarrow$ ~~prime~~

prime = $\{3, 5\}$

$$\Rightarrow \frac{3}{6} = \frac{1}{2} \quad \text{or} \quad \frac{2}{6} = \frac{1}{3}$$

sample space = $\{1, 2, 3, 4, 5, 6\}$.

Let A be the event that the no is odd = $\{1, 3, 5\}$

Let B be the event that its prime $B = \{2, 3, 5\}$

$$P(B|A)$$

prime that's odd

$$= \frac{P(B \cap A)}{P(A)} = \frac{2/6}{3/6} = \frac{1/3}{1/3} = 1$$

* Partition of a set

The subsets of the sets $C_1, C_2, \dots, C_n$ is said to be partition of S, if:

$$1. \bigcup_{i=1}^n C_i = S$$

$$2. C_i \cap C_j = \phi, \quad i \neq j, \quad \forall i, j$$

$$P(A) = \sum_{k=1}^n P(C_k)$$

LAW OF TOTAL PROBABILITY

$C_1, C_2, \dots, C_n \rightarrow$ partitions of sample, S.

* Bayes theorem

$E_1, E_2, \dots, E_n$ set of events associated with sample space, S where all the events $E_1, E_2, \dots, E_n$ have non zero probability of occurrence and they form a partition of S. Let A be any event associated with S, then according to Bayes theorem,

$$P(E_i|A) = \frac{P(E_i) P(A|E_i)}{\sum_{k=1}^n P(E_k)}$$

P.T.O.

Question:

strike $\rightarrow$ ~~0.80~~ 0.65
 construction job = ~~0.32~~ (if there's no strike)
 0.80

Constn job if there's strike = 0.32

soln

A $\rightarrow$ construction job completed on time.

B $\rightarrow$ there will be strike

To find $P(A)$ $\rightarrow$?

Given $P(B) = 0.65$
 $P(B') = 0.35$

$P(A|B') = 0.80$

$P(A|B) = 0.32$

~~$P(A|B)$~~ = ~~0.32~~

According to total probability theorem.

$$P(A) = P(B) \cdot P(A|B) + P(B') \cdot P(A|B')$$

$$P(A) = (0.65 \times 0.32) + (0.35 \times 0.80)$$

$$P(A) = 0.2080 + 0.2800$$

$$P(A) = \underline{\underline{0.4880}}$$

$$\Rightarrow 0.0198$$

$$0.1 \times 0.02$$

$$0.5$$

$$\Rightarrow \frac{1}{10} \times \frac{2}{100} \times \frac{3}{100}$$

$$\Rightarrow \frac{1}{500} \times \frac{1}{5} = \frac{1}{2500}$$

$$\Rightarrow \frac{1}{2500} \times \frac{5}{10} = \frac{1}{5000}$$

Sample Space

$$P(A) = P(B_1) \cdot P(A|B_1) + P(B_2) \cdot P(A|B_2) + P(B_3) \cdot P(A|B_3)$$

$$\frac{1}{65} \times 22$$

$$\frac{130}{10000}$$

$$\frac{1950}{10000}$$

$$\frac{2080}{10000}$$

$$\frac{2080}{10000}$$

2) $X \rightarrow 2Y$
 $X \rightarrow Z$

2% X, 2% Y, 4% Z $\rightarrow$ defective.

suppose the tv chooses in defective, what the probability that this is from company X.

A $\rightarrow$ defective
 $X \rightarrow$ from company X.
 $Y \rightarrow$ from company Y
 $Z \rightarrow$ from company Z

$$\begin{array}{ccc} 0.5 & + & 0.25 \\ \downarrow & & \downarrow \\ X & & Y \\ & & \downarrow \\ & & Z \end{array}$$

$$\left[\begin{array}{l} P(X) = 0.5 \\ P(Y) = 0.25 \\ P(Z) = 0.25 \end{array} \right]$$

$$P(A|X) = P(A_X) = \frac{2}{100} \times 0.5 = \frac{2}{100} \times \frac{50}{100} = \underline{\underline{0.01}}$$

$$P(A|Y) = P(A_Y) = \frac{2}{100} \times 0.25 = \frac{2}{100} \times \frac{25}{100} = \underline{\underline{0.0025}}$$

$$P(A|Z) = P(A_Z) = \frac{4}{100} \times 0.25 = \frac{4}{100} \times \frac{25}{100} = \underline{\underline{0.01}}$$

$$P(A) = P(X) \cdot P(A|X) + P(Y) \cdot P(A|Y) + P(Z) \cdot P(A|Z)$$

$$P(A) = (0.5 \times 0.01) + (0.25 \times 0.0025) + (0.25 \times 0.01)$$

$$X \cdot \left\{ \begin{array}{l} = \left(\frac{5}{10} \times \frac{1}{100} \right) + \left(\frac{25}{100} \times \frac{2}{100} \right) + \left(\frac{25}{100} \times \frac{1}{100} \right) \\ = 0.005 + 0.0005 + 0.00025 \\ \Rightarrow \underline{\underline{0.00575}} \end{array} \right.$$

$$\Rightarrow \underline{\underline{0.25}}$$

$$P(X|D) = \frac{P(X) \cdot P(D|X)}{P(D)} = \frac{0.5 \times 0.02}{0.025} = 0.4$$

$$\boxed{\text{Thus, } P(X|D) = \underline{\underline{0.4}}}$$

Baye's theorem is directly using $P(D)$ eqⁿ in $P(X|D)$

Q1-3) = 3 boxes,
 ① 1 white, 2 red, 3 black
 ② 2 white, 3 red, 1 black
 ③ 3 white, 1 red, 2 black.

A box is chosen at random and from it two balls are drawn at random. One ball is red & the other is white. What is the probability that they come from the second box?

$$\frac{\frac{1}{3} \times \frac{2}{5}}{\frac{1}{3} \times \frac{2}{5} + \frac{1}{3} \times \frac{3}{4}} \Rightarrow \frac{1/15}{1/15 + 1/4} = \frac{1/15}{7/20} = \frac{4}{21}$$

A → second box $P(A) = 1/3$

R → (A|R)

W → ~~(A|R)~~ (A|W)

B → (A|B)

$$P(A|R) = \frac{P(A \cap R)}{P(R)}$$

P

$$P(B_2 | A) = \frac{P(B_2) \cdot P(A|B_2)}{P(A)}$$

$$\text{soln} = 0.0198$$

$$\frac{1}{10} * \frac{99}{100} \quad P(A|B) = 0.5 \quad P(A)$$

$$P(B_1) \rightarrow$$

$$P(B_2) = \underline{\underline{1/3}}$$

$$0.5 * 0.1 + 0.5 * 0.9 \Rightarrow$$

$$P(A) = \underline{\underline{1/2}}$$

$$\frac{5}{10} * \frac{1}{10} \quad \frac{5}{100} + \frac{45}{100} \quad \frac{1}{6} + \frac{1}{6} \Rightarrow \frac{2}{6} = \underline{\underline{1/3}}$$

$$P(A|B_1) = ? \quad \frac{6!}{4! \times 2!} = \frac{6 \times 5}{4 \times 2} = \frac{15}{4}$$

$$P(A|B_2) = ? \quad \frac{1C_1 * 2C_1}{6C_2} =$$

$$P(A|B_1) = \frac{1}{3} \left(\frac{1}{1} + \frac{1}{2} \right) = \frac{3}{2} //$$

$$P(A|B_2) = \frac{1}{3} \left(\frac{1}{2} + \frac{1}{3} \right) = \frac{5}{6} //$$

$$P(A|B_3) = \frac{1}{3} \left(\frac{1}{3} + 1 \right) = \underline{\underline{4/3}}$$

$$\frac{3!}{2!1!}$$

$$2 \times 3 = \underline{\underline{6}}$$

$$\frac{6!}{4! \times 2!}$$

$$\frac{1C_1 * 2C_1}{6C_2} = \frac{1 * \frac{2!}{1!}}{\frac{6!}{4! \times 2!}} \Rightarrow \frac{1}{\underline{\underline{30}}} \Rightarrow \frac{1}{15}$$

$$\frac{2!}{1! \times 1!} * \frac{3!}{2!} \Rightarrow \frac{6}{30} \Rightarrow \frac{1}{5} \quad \frac{6}{15}$$

$$5 \mid 10, 5, 30$$

$$2 \mid 2, 1, 6$$

$$3 \mid 1, 1, 3$$

$$\underline{\underline{30}}$$

$$\frac{3!}{2!} * 1 = \frac{3}{2} \quad \frac{1}{10 \times 3} + \frac{1}{5 \times 6} + \frac{1}{30}$$

$$\Rightarrow \frac{3 + 6 + 1}{30}$$

$$\Rightarrow \frac{10}{30} \times \frac{1}{3} \Rightarrow \underline{\underline{1/9}}$$

$$\frac{1}{15} + \frac{6}{15} + \frac{3}{15} \Rightarrow \frac{10}{15}$$

$$\begin{aligned}
 & 4 \times 99 = 396 \\
 & 0.5 * 0.01 + 0.5 * 0.99 \\
 & \Rightarrow \frac{5}{10} * \frac{1}{100} + \frac{5 * 99}{10 * 100}
 \end{aligned}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\begin{aligned}
 P(B) &= P(A|B_1) \cdot P(B_1) + P(B_2) \cdot P(A|B_2) \\
 &= 0.01 * 0.99 + 0.99 * 0.1 \\
 &\Rightarrow 0.0198
 \end{aligned}$$

$$\begin{array}{r}
 0.0099 \\
 \times 2 \\
 \hline
 0.0198
 \end{array}$$

$$\begin{aligned}
 P(A|B) &= \frac{P(A \cap B)}{P(B)} = \frac{0.0099}{0.0198} \Rightarrow 0.5 \\
 & \quad \downarrow \\
 & \quad 0.5
 \end{aligned}$$

$$\begin{array}{r}
 396 \\
 \hline
 1000 \\
 198 \\
 \hline
 1000
 \end{array}
 \quad
 \begin{array}{r}
 11 \\
 198 \\
 \times 2 \\
 \hline
 396 \\
 0 \\
 \hline
 0.0198
 \end{array}$$

Hence 50% is the answer.

$$P(A|B) = \frac{50}{100} = 0.5 \text{ (50\%)}$$

10/10/25
Friday

ONE DIMENSIONAL RANDOM VARIABLES

Sample space is difficult to describe if elements are not numbers. Now we formulate results to represent sample space as numbers.

RANDOM VARIABLE — a variable present in the sample space of a random experiment

Consider random exp of tossing a die
 $X: S \rightarrow R$ be a function defined by
 $X(s) = 10s$

Let $X: S \rightarrow R$ be a fn. defined by

$$X(s) = \begin{cases} 0 & \text{if } s \text{ is odd} \\ 1 & \text{if } s \text{ is even} \end{cases}$$

Continuous
 $\downarrow$
 Integration has to be done.

Discrete
 $\downarrow$
 Probability for each

Qn) $x =$ number of heads in 3 coin tosses

$$S = \left\{ \begin{array}{ccccccc} \text{HHH}, & \text{HTH}, & \text{HHT}, & \text{THH}, & \text{HTT}, & \text{TTH}, & \text{THT}, & \text{TTT} \\ 3 & 2 & 2 & 2 & 1 & 1 & 1 & 0 \end{array} \right\}$$

$$x \Rightarrow \text{no of heads.}$$

$$X = \{0, 1, 2, 3\}$$

probability to each sample space = $1/8$

$$P_1 = P(X=0) = \text{TTT} = 1/8$$

$$P_2 = P(X=1) = 3/8$$

$$P_3 = P(X=2) = 3/8$$

$$P_4 = P(X=3) = 1/8$$

$$\text{Hence } \boxed{P = \sum_{i=1}^4 P_i = 1}$$

Qn) continuous case $f(x) = \begin{cases} 2x & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$ verify if a

Integration pdf & find $P(X < 1/2)$.
For continuous x : has density $f(x)$, $\int f(x) dx = 1$

$$\int_0^1 2x dx \Rightarrow \left[\frac{2x^2}{2} \right]_0^1$$

$$\Rightarrow 1^2 - 0 = 1$$

$$\text{Qn} \quad P(X < 1/2)$$

$$(i) f(x) \geq 0$$

$$\text{If } x < 0, f(x) = 0$$

$$\text{If } 0 < x < 1, f(x) = 2x \geq 0$$

$$\text{If } x > 1, f(x) = 0$$

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^{\infty} f(x) dx$$

$$= \int_{-\infty}^0 0 dx + \int_0^1 2x dx + \int_1^{\infty} 0 dx$$

$\Rightarrow \underline{\underline{1}}$ hence f is a pdf.

Given $f(x) < 1/2$
 In this case we will be considering all the values below or less than $1/2$

$$\Rightarrow \int_{-\infty}^0 f(x) dx + \int_0^{1/2} f(x) dx$$

$$\Rightarrow \int_{-\infty}^0 0 dx + \int_0^{1/2} 2x dx$$

$$\Rightarrow \left[\frac{x^2}{2} \right]_0^{1/2} = \frac{(1/2)^2 - 0^2}{2} = \underline{\underline{1/4}}$$

Expectation

$$E[x] = \begin{cases} \sum x_i p_i, & \text{if } x \text{ is discrete} \\ \int x f(x) dx, & \text{if } x \text{ is continuous} \end{cases}$$

Moment Generating Function, (MGF)

$$M_x(t) = E[e^{tx}] = \begin{cases} \sum e^{tx_i} p_i, & \text{if } x \text{ is discrete} \\ \int e^{tx} f(x) dx, & \text{if } x \text{ is continuous} \end{cases}$$

Moments: $E[x^r] = M_x^{(r)}(0)$

$\downarrow$
 { Differentiate M_x r times at $t=0$ }

question: Discrete $P(x=0) = P(x=1) = 1/2$

Given $P_1 = 1/2$ $P_2 = 1/2$

~~$\sum x_i$~~ ~~P_i~~

$$\sum x_i p_i \Rightarrow x_1 p_1 + x_2 p_2$$

$$\Rightarrow (0 \times 1/2) + (1 \times 1/2) = \underline{\underline{1/2}}$$

~~Question~~ Moment Generating function

$$M_x(t) = E[e^{tx}] = \sum_1^n e^{tx_i} \cdot p_i$$

$$\frac{d}{dt} \left(\frac{1+et}{2} \right) \Rightarrow \frac{0+et}{2} = \left[\frac{et}{2} \right]_{t=0}$$

$$\Rightarrow \frac{e^0}{2} = \underline{\underline{\frac{1}{2}}}$$

$$e^{tx_1} p_1 + e^{tx_2} p_2$$

$$\frac{e^{t(0)}}{2} + e^{t(1)} \cdot 1/2$$

$$\frac{1}{2} (1 + et) \Rightarrow \underline{\underline{\left(\frac{1+et}{2} \right)}}$$

Question for continuous

$$f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$0 < x < 1$

otherwise

$$\int_{-\infty}^{\infty} f(x) = \int_{-\infty}^0 0 + \int_0^1 2x + \int_1^{\infty} 0 = \underline{\underline{1}}$$

Moment Generating

$$\int e^{tx} \cdot 2x \Rightarrow \int_u^v e^{tx} \cdot x \Rightarrow$$

$$\int u v = \int u \int v - \int du \int v \quad \text{here } v = e^{tx}$$

$$u = x$$

$$= x \left(\frac{e^{tx}}{t} \right) - \int 1 \times \frac{e^{tx}}{t}$$

$$MGF \Rightarrow x \left(\frac{e^{tx}}{t} \right) - \frac{e^{tx}}{t^2} \Rightarrow \underline{\underline{\frac{e^{tx}}{t} \left(x - \frac{1}{t} \right)}}$$

$$E(x) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^1 x (2x) dx = 2 \int_0^1 x^2 dx$$

$$\Rightarrow 2 \left[\frac{x^3}{3} \right]_0^1 \Rightarrow 2 \left[\frac{1}{3} - 0 \right] \Rightarrow \frac{2}{3}$$

$$M_x(t) = \int_{-\infty}^{\infty} e^{tx} \cdot f(x) dx \Rightarrow 2 \int_0^1 x e^{tx} dx$$

$$2 \left[x \cdot \left(\frac{e^{tx}}{t} \right) - \int 1 \cdot \frac{e^{tx}}{t} \right]_0^1$$

$$2 \left[x \frac{e^{tx}}{t} - \frac{e^{tx}}{t^2} \right]_0^1$$

$$2 \left[- \left[\cancel{0 \cdot \frac{e^{t0}}{t}} - \frac{e^{t0}}{t^2} \right] + \left[\frac{1 \cdot e^t}{t} - \frac{e^t}{t^2} \right] \right]$$

$$2 \left[+ \frac{1}{t^2} + \frac{e^t}{t} - \frac{e^t}{t^2} \right]$$

$$\Rightarrow \boxed{\frac{2}{t} \left[-\frac{e^t}{t} + e^t + \frac{1}{t} \right]}$$

* probability mass fn.
 $\left. \begin{aligned} & p_i \geq 0 \text{ for all } i \\ & \sum_{i=1}^n p_i = 1 \end{aligned} \right\} \leftarrow \text{satisfies condition.}$

* probability density fn.
 $\left. \begin{aligned} & f(x) \geq 0 \text{ for all } x \\ & \int_{-\infty}^{\infty} f(x) dx = 1 \end{aligned} \right\} \leftarrow \text{satisfies conditions}$

* Distribution function
 $\rightarrow$ also called as cumulative distribution function,
 of a random variable x is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt.$$
 $\downarrow$
 gives total probability accumulated to the point x .

* properties of distribution function
 $\rightarrow F(x)$ is non decreasing
 $\rightarrow \lim_{x \rightarrow -\infty} f(x) = 0$ and $\lim_{x \rightarrow \infty} f(x) = 1$
 $\rightarrow P(X > x) = 1 - F(x)$

1.) If x is a random variable with pdf

$$f(x) = \begin{cases} e^{-x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the distribution function.

Soln:-

$$D.F = P(x) = \int_{-\infty}^x f(t) dt$$

$$\Rightarrow \int_{-\infty}^0 f(t) dt + \int_0^x f(t) dt \Rightarrow \int_0^x e^{-t} dt$$

$$\left[\frac{e^{-t}}{-1} \right]_0^x$$

$$\Rightarrow - \left(e^{-t} \right)_0^x \Rightarrow -1(e^{-x} - e^0)$$

$$\Rightarrow -(e^{-x} - 1)$$

$$\Rightarrow \boxed{1 - e^{-x}}$$

Soln

$$F(x) = \begin{cases} 1 - e^{-x} & \text{if } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

2) If x is a random variable with pmf

$$p(x) = \begin{cases} \frac{x}{15} & x = 1, 2, 3, 4, 5 \\ 0 & \text{otherwise} \end{cases}$$

Find a) $P(x=1 \text{ or } x=2)$.

~~$P(x=1) \text{ or } (x=2)$~~

b) $P\left(\frac{1}{2} < x < \frac{5}{2} \mid x > 1\right)$.

(a) $P(x=1 \text{ or } x=2)$

$$P(x=1) + P(x=2)$$

$$\frac{1}{15} + \frac{2}{15} = \frac{3}{15} = \underline{\underline{\frac{1}{5}}}$$

$$(b) P\left(\frac{1}{2} < x < \frac{5}{2} \mid x > 1\right).$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\text{thus: } \frac{P\left(\frac{1}{2} < x < \frac{5}{2}\right) \cap P(x > 1)}{P(x > 1)}$$

$$5/2 \Rightarrow 2.5$$

$$1/2$$

$$x > 1 \Rightarrow 2, 3, 4, 5$$

$$\Rightarrow \frac{P(\{2, 3, 4, 5\})}{P(\{2, 3, 4, 5\})} = \frac{1}{4} = \frac{1}{4}$$

4 elements

$$\Rightarrow \frac{P(2)}{P(2) + P(3) + P(4) + P(5)} = \frac{2/15}{2+3+4+5} = \frac{2}{14} = \frac{1}{7}$$

Question 3). If x is a crv with pdf $f(x) = \begin{cases} c(4x - 2x^2), & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$

Find (i) c

(ii) $P(x > 1)$

(iii) distribution function

$$\text{Soln: } \int_{-\infty}^{\infty} f(x) dx = 1$$

(probability density fn)

$$\int_0^2 c(4x - 2x^2) dx = 1.$$

$$c \left[\frac{4x^2}{2} - \frac{2x^3}{3} \right]_0^2 = 1$$

$$\left[2(2^2) - \frac{2(2^3)}{3} \right] - 0 = \frac{1}{c}$$

$$\Rightarrow 8 - \frac{16}{3} = \frac{24-16}{3} = \frac{8}{3}$$

$$\boxed{c = 3/8}$$

$$(ii) P(x > 1).$$

$$\Rightarrow 1 - P(x \leq 1)$$

$$\rightarrow 1 - \int_{-\infty}^1 f(x) \cdot dx$$

$$1 - \int_0^1 f(x) \cdot dx.$$

$$1 - \int_0^1 c \cdot (4x - 2x^2) dx$$

$$1 - c \left[\frac{4x^2}{2} - \frac{2x^3}{3} \right]_0^1$$

$$1 - \frac{3}{8} [2(1) - 2(1)]$$

$$\Rightarrow \underline{\underline{1}}$$

$$1 - \frac{3}{8} \left[2 - \frac{2}{3} \right]$$

$$1 - \frac{3}{8} \left[\frac{6-2}{3} \right]$$

$$(iii) \text{ distribution function: } 1 - \frac{3}{8} \left(\frac{4}{3} \right) \Rightarrow \underline{\underline{1/2}}$$

$$\int_{-\infty}^x f(t) \cdot dt$$

$$\int_0^x c (4t - 2t^2) \cdot dt$$

$$\left. \begin{aligned} & \int_0^2 \frac{3}{8} (4x - 2x^2) \\ & \Rightarrow \frac{3}{8} \left(\frac{4x^2}{2} - \frac{2x^3}{3} \right) \\ & \Rightarrow \frac{3}{8} \left[\left(\frac{2^3 - 2^4}{3} \right) - \left(\frac{0 - 0}{3} \right) \right] \\ & \Rightarrow \frac{3}{8} \left[\left(\frac{8 - 16}{3} \right) - \left(\frac{0}{3} \right) \right] \\ & \Rightarrow \frac{3}{8} \left[\frac{8}{3} - \frac{4}{3} \right] \\ & \Rightarrow \frac{4}{8} \end{aligned} \right\}$$

$$C \int_0^x \left(\frac{4t^2}{2} - \frac{2t^3}{3} \right) dt.$$

$$\Rightarrow C \left[2x^2 - \frac{2x^3}{3} \right]$$

$$\Rightarrow \frac{3}{8} \left[2x^2 - \frac{2x^3}{3} \right] \} \rightarrow \text{finite integral, so no use of adding constant after integration.}$$

Thus $F(x) = \begin{cases} 0 & \text{if } x \leq 0. \\ \frac{3}{8} \left(2x^2 - \frac{2}{3} x^3 \right) & \text{if } 0 < x < 2 \\ & \text{if } x \geq 0 \end{cases}$

$$\int_2^x C(4t - 2t^2) \cdot dt$$

$$\Rightarrow \left[\frac{4t^2}{2} - \frac{2t^3}{3} \right]_2^x$$

$$\Rightarrow \frac{3}{8} \left(2x^2 - \frac{2x^3}{3} \right) - \left(2(2^2) - \frac{2}{3} 4 \right)$$

$$\frac{3}{8} \left(2x^2 - \frac{2x^3}{3} \right) - 1$$

Hence $\Rightarrow \underline{\underline{1}}$

Qn) H.W The diameter of an electric cable is assumed to be

crv with pdf

$$f(x) = \begin{cases} kx^3(1-x), & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

soln

Find

(a) k

$\longrightarrow 20$

(b) $P(X < 2/3)$

$\longrightarrow 0.4609$ or $\frac{112}{243}$

(c) distribution function

$$F(x) = \begin{cases} 0 & x < 0 \\ 5x^4 - 4x^5 & 0 < x < 1 \\ 1 & x \geq 1 \end{cases}$$

Qn) A crv has pdf $f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

Find $P(X < 1/2) \longrightarrow \text{soln } 1/4$

$P(X > \frac{3}{4} \mid X > \frac{1}{2}) = 7/12$

$P(X < 3/4 \mid X > 1/2) = 5/12$

$\rightarrow$ Hw 1)

$$\int_0^1 kx^3(1-x) = 1$$

$$k \int_0^1 x^3 - x^4 = 1$$

$$k \left[\frac{x^4}{4} - \frac{x^5}{5} \right]_0^1 = 1$$

$$\left[\frac{1}{4} - \frac{1}{5} \right] = \frac{1}{k}$$

$$\frac{5-4}{20} = \frac{1}{k}$$

$$\boxed{k = 20}$$

$$b) P(x < 2/3) \\ \int_0^{2/3} K [x^3 - x^4]$$

$$20 \int_0^{2/3} x^3 - x^4 = 20 \left[\frac{x^4}{4} - \frac{x^5}{5} \right]_0^{2/3}$$

$$\Rightarrow 20 \left(\frac{(2/3)^4}{4} - \frac{(2/3)^5}{5} \right)$$

$$\Rightarrow 20 \left(\frac{2^4}{3^4 \cdot 4} - \frac{2^5}{3^5 \cdot 5} \right)$$

$$20 \left(\frac{16 \times 3}{81 \times 3} - \frac{32}{243} \right) = \frac{48 - 32}{243}$$

$$\begin{array}{r} 3 \overline{) 81, 243} \\ 9 \overline{) 27, 81} \\ 3 \overline{) 9, 27} \\ 3 \overline{) 3, 9} \\ 1, 1 \end{array}$$

$$3 \times 3 \times 3 \times 3$$

$$\frac{27}{\times 3}$$

$$\frac{81}{\times 3}$$

$$\Rightarrow \frac{16 \times 20}{243} = \frac{320}{243}$$

16

MATHEMATICS

14/10/25
Tuesday

$$\int_{3/4}^1 2x$$

$$x > 3/4$$

$$\Rightarrow \left[\frac{2x^2}{2} \right]_{3/4}^1$$

$$P(x > 1/2)$$

$$\int_{1/2}^1 x^2 = 1^2 - \left(\frac{1}{2}\right)^2 = 1 - \frac{1}{4} = \frac{3}{4}$$

$$1^2 - \left(\frac{3}{4}\right)^2 = 1 - \frac{9}{16} = \frac{16-9}{16} = \frac{7}{16}$$

Hence $\frac{P(x > 3/4)}{P(x > 1/2)}$

$$\Rightarrow \frac{7/16}{3/4} = \frac{7}{12} \quad \times$$

$$\frac{P(x > 3/4)}{P(x > 1/2)} = \frac{1 - P(x \leq 3/4)}{1 - P(x \leq 1/2)}$$

$$\Rightarrow \frac{1 - \int_0^{3/4} x^2}{1 - \int_0^{1/2} x^2} = \frac{1 - \frac{9}{16}}{1 - \frac{1}{4}}$$

$$\frac{7}{12} < \frac{\frac{16-9}{16}}{3/4} = \frac{7/16}{3/4}$$

c) $P(x < 3/4 | x > 1/2)$

$$\Rightarrow \frac{\int_{1/2}^{3/4} x^2}{1 - \int_0^{1/2} x^2}$$

$$\frac{9/16}{1 - (1/2)^2}$$

$$\Rightarrow \frac{9/16}{1 - 1/4} = \frac{9/16}{3/4} = 3/4 \quad \times$$

$$\frac{P(x < 3/4) \cap (x > 1/2)}{P(x > 1/2)}$$

$$= \frac{P(1/2 < x < 3/4)}{P(x > 1/2)}$$

probability distribution
↓
pdf

$$\frac{\int_{1/2}^{3/4} x^2 dx}{\int_{1/2}^1 x^2 dx} \Rightarrow$$

$$\frac{\frac{9}{16} - \frac{1}{4} \times \frac{1}{4}}{1 - \frac{1}{4}} = \frac{5/16}{3/4} = \frac{5}{12}$$

Solution is $\boxed{P(x < \frac{3}{4} | x > \frac{1}{2})} \Rightarrow \underline{\underline{\frac{5}{12}}}$

*. Two dimensional Random variables
2 types → discrete & continuous

S → sample space.

$$x = X(\omega)$$

$$y = Y(\omega) \quad \omega \in S$$

{ If x & y are finite → discrete
If infinite → these are continuous }

*. Joint probability distribution

discrete { satisfies the following conditions :-
① $P(x_i, y_j) \geq 0$ for all (x, y)
② $\sum_i \sum_j P(x_i, y_j) = 1$

continuous { ① $f(x, y) \geq 0$ for all R .
② $\iint_R f(x, y) dx dy = 1$

Distribution function
for random variable (X, Y) is

$$F(X, Y) = P(X \leq x, Y \leq y)$$

$$\Rightarrow \int_{-\infty}^x \int_{-\infty}^y f(x, y) dx dy.$$

$$P(X \leq x) = \int f(x) dx.$$

both the integrals
starts from the
lowest negative
value.

* Marginal Probabilities

→ describe individual behaviour of X or Y .

Discrete
(for X) $P_X(x) = \sum_y P(X, y)$

$$P_Y(y) = \sum_x P(x, y)$$

continuous $f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$

* Conditional Distributions :

Conditional probabilities describe the distribution of one variable given that the other is fixed.

Discrete: $P(X = x | Y = y) = \frac{P(X, y)}{P_Y(y)}$

$$P(Y = y | X = x) = \frac{P(X, y)}{P_X(x)}$$

continuous:

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} ; f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)}$$

Qn:- Suppose 2 ~~cards~~ cards are drawn at random from a pack of cards. Let X be the number of aces and Y be the number of queens obtained

- find joint probability distribution of (X, Y)
- find marginal pdf of X and Y .
- Find $P(X \leq 1, Y > 1)$
- Find $P(X + Y > 1)$
- Find conditional probability distribution function of X given $Y = 1$.
- Find conditional probability distribution of Y given $X = 1$

Soln:- Total no. of cards = 52.
 No. of aces = 4 (X)
 No. of queens = 4 (Y)

sample size

from a standard deck of cards, 2 cards are drawn at random, hence the sample size is

$$= \binom{52}{2} = {}^{52}C_2 = \frac{52!}{50! \times 2!} = \frac{26 \times 51}{2}$$

$$\begin{array}{r} 51 \\ \times 26 \\ \hline 306 \end{array}$$

$$\begin{array}{r} 1020 \\ + 306 \\ \hline 1326 \end{array}$$

(a) Joint probability distribution $\Rightarrow 1326$

* X : no. of aces $\{0, 1, 2\}$

* Y : no. of queens $\{0, 1, 2\}$

$$X + Y \leq 2 = \{(0,0), (0,1), (1,0), (1,1), (2,0), (0,2)\}$$

$$P(X=0, Y=0) = \frac{4C_0 \times 4C_0 \times 44C_2}{52C_2}$$

$$\Rightarrow \frac{4 \times 4 \times \frac{44 \times 43}{2}}{52 \times 51 / 2} = \frac{4 \times 946}{1326}$$

$$\begin{array}{r} 43 \\ \times 22 \\ \hline 86 \\ 866 \\ \hline 946 \end{array}$$

$$\frac{4!}{3! \times 0!}$$

$$\begin{array}{r} 33 \\ 4 \overline{)1326} \\ \underline{12} \\ 12 \end{array}$$

$$\Rightarrow \frac{473}{663}$$

$$\Rightarrow \frac{946}{1326}$$

$$\frac{44 \times 4}{6}$$

$$P(X=0, Y=1) = \frac{4C_0 \times 4C_1 \times 44C_1}{52C_2} = \frac{4 \times 44}{1326} = \frac{88}{663}$$

$$P(X=1, Y=0) = \frac{4C_1 \times 4C_0 \times 44C_1}{52C_2} = \frac{88}{663}$$

$$P(X=1, Y=1) = \frac{4C_1 \times 4C_1 \times 44C_2}{52C_2} = \frac{4 \times 4 \times \frac{4!}{2! \times 2!}}{1326} = \frac{8}{663}$$

$$\begin{array}{r} 1 \\ 44 \times 43 \\ \hline 132 \\ 1760 \\ \hline 1892 \end{array}$$

$$6 \overline{)1326} \\ \underline{12} \\ 12 \\ \underline{12} \\ 06$$

$$P(X=0, Y=2) = \frac{4C_0 \times 4C_2 \times 44C_2}{52C_2} = \frac{6 \times 1326}{1326 \times 261}$$

$$P(X=2, Y=0) = \frac{1}{221}$$

X \ Y	0	1	2
0	$\frac{473}{663}$	$\frac{88}{663}$	$\frac{1}{221}$
1	$\frac{88}{663}$	$\frac{8}{663}$	0
2	$\frac{1}{221}$	0	0

Math

16/10/25

$x \backslash y$	0	1	2	$P(x)$	
0	$473/663$	$88/663$	$1/221$	$188/221$	$(x=0)$
1	$88/663$	$8/663$	0	$32/221$	$(x=1)$
2	$1/221$	0	0	$1/221$	$(x=2)$
$P(y)$	$188/221$	$32/221$	$1/221$		
	$y=0$	$y=1$	$y=2$		

marginal probability for x values & y being constant.

marginal probability for y values & x being constant

c) $P(x \leq 1, y > 1)$
 $\Rightarrow$ ~~$P(0,2)$~~ ~~$P(1,2)$~~
 $x = 0, 1$
 $y = 2$
 $P(0,2) + P(1,2)$
 $1/221 + 0 = 1/221$

d) $P(x+y > 1)$
 $(0,2), (2,0), (1,1), (1,2), (2,1), (2,2)$
 $\Rightarrow \frac{8}{663} + \frac{1}{221} + \frac{1}{221} = \frac{14}{663}$

$P(x|y=1) = \frac{P(x = \{0,1,2\} \cap y = \{1\})}{P(y=1)}$

$\Rightarrow \frac{P(0,1)}{32/221}$

$$\Rightarrow \begin{array}{c|ccc} x: & 0 & 1 & 2 \\ \hline y=1 & 8/663/P(Y=1) & 8/663/P(Y=1) & 0 \end{array}$$

$$\Rightarrow \frac{8/663}{32/221} - \frac{8/663}{32/221}, \quad 0/32/221$$

$$\Rightarrow \frac{88}{96} \quad 8/96 \quad 0$$

final soln

$x:$	0	1	2
$y=1$	$88/96$	$8/96$	0

$$f) P(Y|X) = \frac{P(Y, X)}{P(X)} = \frac{P(X, Y)}{P(X)}$$

$x \backslash y$	0	1	2	$P(X)$
0	$473/663$	$88/663$	$1/221$	$188/221$
1	$88/663$	$8/663$	0	$32/221$
2	$1/221$	0	0	$1/221$
	$188/221$	$32/221$	$1/221$	

$\Rightarrow$

0

X/Y	0	1	2
0	$\frac{1173/663}{188/221}$	$\frac{88/663}{188/221}$	$\frac{1/221}{188/221}$
1	$\frac{88/663}{32/221}$	$\frac{8/663}{32/221}$	$\frac{0}{32/221}$
2	$\frac{1/221}{1/221}$	$\frac{0}{0}$	$\frac{0}{0}$

	0	1	2
0	$\frac{413}{564}$	$\frac{22}{47}$	$\frac{1}{188}$
1	$\frac{11}{12}$	$\frac{1}{12}$	0
2	1	0	0

Formulae & Variables :- Variance equations.

Expectation

$$E[X] = \sum_i x_i P_X(x_i)$$

$$E[Y] = \sum_i y_i P_Y(y_i)$$

Variance

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

standard deviation

$$S.D = \sqrt{\text{Variance}}$$

Covariance

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

correlation

coefficient:-

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}$$

$$-1 \leq \rho_{XY} \leq 1$$

$\rho = +1 \rightarrow$ perfect positive linear relationship
 $\rho = -1 \rightarrow$ perfect negative linear relationship
 $\rho = 0 \rightarrow$ no relationship

$|\rho| < 0.3$: ~~mod~~ poor correlation
 $0.3 \leq |\rho| \leq 0.7$: moderate correlation
 $|\rho| > 0.7$: strong correlation

Qn:- 1.)

	x^2	0	1	
y^2	$x \backslash y$	0	1	
0	$y=0$	0.1	0.2	$P(X=0) \Rightarrow 0.3$
1	$y=1$	0.3	0.4	$P(X=1) \Rightarrow 0.7$
		$P(Y=0)$	$P(Y=1)$	
		0.4	0.6	

$\frac{3}{10} + \frac{3}{10} \Rightarrow 0.09$

$0.4 \times 1 \times 1$

- Expectations of x & y
- Variations of x & y
- Std of x & y
- Covariance
- Correlation coefficient

Soln:-

- Expectations of x & y

$$E[x] = (0.1 \times 0.3 + 0.2 \times 0.3)$$

$$E[x] = 0 \times P(X=0) + 1 \times P(X=1)$$

$$= 0 \times 0.3 + 1 \times 0.7 \Rightarrow \underline{0.7}$$

$$E[y] = 0 \times 0.4 + 1 \times 0.6 = \underline{0.6}$$

- Variations of x & y

$$\text{Var}(x) = E[x^2] - (E[x])^2$$

$$\text{Var}(x) = E[x^2] = 0 \times 0.3 + 1 \times 0.7 = 0.7$$

$$\Rightarrow \text{Var}(x) = 0.7 - (0.7)^2 \Rightarrow$$

$$\text{var } y = E[y^2] - (E[y])^2$$

$$\text{var } (x) = E[x^2] - (E[x])^2$$

$$E[x^2] = 0.7$$

$$E[y^2] = 0.6$$

$$\Rightarrow \text{var}(x) = 0.7 - 0.49 = 0.21$$

$$\text{var}(y) = 0.24$$

c) std of x & y

$$\text{std } x = \sqrt{\text{var } x} = \sqrt{0.21} = 0.458$$

$$\text{std } y = \sqrt{\text{var } y} = \sqrt{0.24} = 0.490$$

d) Covariance:-

$$E[x] = 0.7$$

$$E[y] = 0.6$$

$$E[xy] - E[x]E[y]$$

$$E[xy] = 1 \times 1 \times P(x=1, y=1)$$

$$E[xy] = x_i y_i P(x=0) + x_i y_i P(x=1)$$

$$1.3 - (0.7 \times 0.6) = 1.3 - 0.42 = 0.88$$

$$0.4 - (0.7 \times 0.6) = 0.4 - 0.42 = -0.02$$

e) correlation coefficient

$$\rho(x, y) = \frac{\text{Cov}(x, y)}{\sqrt{\text{var}(x) \text{var}(y)}} = \frac{-0.02}{\sqrt{0.21 \times 0.24}}$$

(perfect negative linear relationship) -ve value soln

* Generating function:

$a_1, a_2, \dots, a_n \rightarrow$ sequence of real numbers.
 $A(s) = a_0 + a_1s + a_2s^2 + \dots = \sum_{k=0}^{\infty} a_k s^k$

Interval $-s_0 < s < s_0$, then $A(s)$ is called a generating function of sequence $a_0, a_1, a_2, \dots$

* Probability Generating function:-

Suppose that X is a random variable which assumes non-negative ~~integer~~ integral values $0, 1, 2, \dots$ and $P(X=x) = p_x, (x=0, 1, 2, \dots)$.

Probability mass function of Bernoulli distribution.

$$P(X) = p^x (1-p)^{1-x}$$

Mean / ~~Exp~~ Expectation = P . (Mean = $E[X]$). Mean & expectation = P

$$E[X^2] = P$$

$$\text{Variance } [N(X)] = P - P^2$$

$$\Rightarrow \text{Variance } P(1-P) = P \times q \quad (1-P=q)$$

Qn) A machine produces items & each item is defective with probability 0.1 and non-defective with probability 0.9.

- define Bernoulli's, x.v for item is defective
- find pmf, expectation & variance

non-defective $\rightarrow 0.9$

defective $\rightarrow 0.1$

Soln:-

$$a) X(\text{defective}) = ?$$

$$X(\text{defective}) = 1$$

$$X(\text{non defective}) = 0$$

$$b) \text{Pmf} \Rightarrow p^x (1-p)^{1-x}$$

$$\Rightarrow 0.1^x (1-0.1)^{1-x} \Rightarrow (0.1)^x (0.9)^{1-x}$$

$$E[X]: \text{expectation} = P = 0.1$$

$$\text{Variance} = pq = 0.1 \times 0.9 \Rightarrow 0.09$$

$$\frac{1}{10} \times \frac{9}{10} \rightarrow$$

Question (Q.) A student guesses the ans to a true/false qn.
 a) Define Bernoulli's r.v. X for answer is correct
 b) find pmf, expectation & variance

Soln:-

a)

$X(\text{correct}) = 1$	(0.5)
$X(\text{false wrong}) = 0$	(0.5)

Since its true/false question $\rightarrow$ probability for each would be 0.5

b) pmf: $p^x (1-p)^{1-x}$

$$\Rightarrow (0.5)^x (1-0.5)^{1-x} = (0.5)^x \cdot (0.5)^{1-x}$$

pmf $\Rightarrow 0.5^{x+1-x} = 0.5$

expectation

$$E[X] = p = 0.5$$

Variance $\therefore = pq = 0.5 * 0.5 = 0.25$

* BINOMIAL DISTRIBUTION.

$\rightarrow$ If there exists 2 variables n and p .
 X = number of successes in n independent Bernoulli trials with success prob p , then

$$P(X=x) = \binom{n}{x} p^x q^{n-x}, \quad x=0,1,\dots,n$$

$$p+q=1.$$

$n \rightarrow$ no of trials

$p \rightarrow$ success.

$$q = 1-p \text{ (failure)}$$

$x \rightarrow$ no. of success, in n trials.

* PMF

$$p(x) = \begin{cases} \binom{n}{x} p^x q^{n-x} & \text{if } x=0,1,\dots,n \\ 0 & \text{otherwise} \end{cases}$$

pgf $\rightarrow$ probability generating function:

Then $p(s) = (q + ps)^n$

pmf $\rightarrow$ probability mass function
pgf $\rightarrow$ probability generating function.

Mean and Variance

$\downarrow$
expectation $E(x) = np$.

$$\boxed{V(x) = npq} \quad \text{Variance}$$

pgf $\rightarrow$ probability generating function.

Qn) = Disks produced by a company will be defective with probability 0.01. The company sells the disks in a packet of 10 & offers money back guarantee if more than one defective disk found in a packet.
What is the proportion of packages returned?
If someone buys 3 packets, what is the probability that exactly one of them should be replaced.

Soln:

Given $p = 0.01$
 $n = 10$.

$X \sim \text{binomial } (n=10, p=0.01)$

$$P(X > 1) = 1 - [P(X=0) + P(X=1)]$$

$$\downarrow$$
$$1 - P(X \leq 1) \quad \boxed{1 - P(X \leq 1)}$$

$$\boxed{P(X=x) = \binom{n}{x} p^x q^{n-x}}$$

Given $x = 0$.

$$P(X=0) = \binom{n}{0} p^0 \cdot q^{n-0}$$

given $n = 10$.

there, $P(X=0) = \binom{10}{0} p^0 \cdot q^{10-0}$

$$\rightarrow \binom{10}{0} p^0 \cdot q^{10}$$

given $p = 0.01$
 $(1-p) q = 0.99$
 $\Rightarrow 0.90438$

(condition: 2)
 $n = 3$
 $p(x) \rightarrow$ probability
 packets returned.

$$P(X=2) = \binom{n}{x} p^x q^{n-x}$$

$$P(X=1) = \binom{10}{1} p^1 q^{10-1}$$

$$\binom{10}{1} (0.01)^1 (0.99)^9 \Rightarrow \underline{0.09135}$$

Also: $1 - [P(X=0) + P(X=1)]$

$$\Rightarrow 1 - (0.90438 + 0.09135)$$

$$\Rightarrow \underline{0.00427}$$

Let Y be the event packet returned = Success.

$$P_Y = 0.00427$$

$n = 3$ (because 'someone buys 3 packets')

$Y \sim (\text{binomial})$

$\xrightarrow{n=3, P_Y = 0.00427}$ only for the sub question

$$P(Y=1) = ?$$

$$P(Y=1) = \binom{n}{y} p^y q^{n-y}$$

$$P(Y=1) = \binom{3}{1} (0.00427)^1 (1-0.00427)^2$$

Hence $P(Y=1) \Rightarrow \underline{0.0127}:-$

23/10/25.

MATH.

NORMAL DISTRIBUTION + (GAUSSIAN DISTRIBUTION).

$$X \sim N(\mu, \sigma^2)$$

$\mu \rightarrow$ mean
 $\sigma^2 \rightarrow$ variance

bell shaped
and
symmetric
about
the mean.

Bell shaped representation.



$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} ; -\infty < x < \infty$$

Symmetric about $x = \mu$.

$x_1, x_2, \dots, x_n \rightarrow$ independent variables
 Sum, $S = x_1 + x_2 + \dots + x_n$.

$$E[S] = E[x_1] + E[x_2] + \dots + E[x_n]$$

$$V[S] = V[x_1] + V[x_2] + \dots + V[x_n]$$

normally distributed $Q = a_1 x_1 + a_2 x_2 + \dots + a_n x_n + b$.

$$E[Q] = a_1 E[x_1] + a_2 E[x_2] + \dots + a_n E[x_n] + b$$

$$V[Q] = a_1^2 V[x_1] + a_2^2 V[x_2] + \dots + a_n^2 V[x_n]$$

$$E[Q] = (1 \times 12) + (-2 \times -5) + (8 \times 3) + (10 \times -4)$$

$$= 12 + 10 + 24 - 40 = 6$$

$$V[Q] = (1 \times 4) + (4 \times 2) + (8 \times 5) + (16 \times 1)$$

$$= 4 + 8 + 40 + 16 = 68$$

$$sd = \sqrt{68} = 8.246$$

Qn) 1.]

	means	variance
Bolt	312.8	2.67
Nut	53.2	0.85
washer	17.5	0.21

Find the distribution of the weight of the assembly.
Find the mean, variance & standard deviation.

$$S = x_1 + x_2 + \dots + x_n$$

$$S = x_1 + x_2 + x_3$$

The distribution of weight of the assembly is normal because weights of bolts, nuts & washers are not dependent to each other.

$$\text{Mean} = E[x_1] + E[x_2] + E[x_3]$$

$$= 312.8 + 53.2 + 17.5 = 383.5$$

(Mean weight of the assembly)

$$\begin{array}{r} 11 \\ 312.8 \\ 53.2 \\ 17.5 \\ \hline 383.5 \end{array}$$

$$V = 2.67 + 0.85 + 0.21$$

$$V = 3.73$$

$$s.d = \sqrt{V} = \sqrt{3.73} = 1.92$$

$$\begin{array}{r} 11 \\ 2.67 \\ 0.21 \\ 0.85 \\ \hline 3.73 \end{array}$$

Independent random variable x_1, x_2, x_3, x_4

	means	variance
$x_1 \rightarrow$	12	4
$x_2 \rightarrow$	-5	2
$x_3 \rightarrow$	8	5
$x_4 \rightarrow$	10	1

$$Q = x_1 - 2x_2 + 3x_3 - 4x_4 + 5$$

$$a_1 = 1$$

$$a_2 = -2$$

$$a_3 = 3$$

$$a_4 = -4$$

$$12 + 10 + 24 - 40$$

$$\Rightarrow 46 - 40 \Rightarrow 6$$

$$\frac{6}{58}$$

$$\begin{array}{r} 4+8+16 \\ 28+45 \\ \hline \Rightarrow 73 \end{array}$$



STATISTICS:-

- Population:- entire collection of individuals/items under study
- Sample:- subset of population selected for analysis
- Sampling:- choosing from a sample.

Need for sampling:-

cost effective
Time saving
feasibility
Accuracy

Basic terms

- Sampling unit.
- Sampling frame.
- parameter $\rightarrow \mu, \text{variance, sd}$
- Sampling error.

1. Simple random sampling $\rightarrow$ eg:- one student out of 20 students
2. Systematic Sampling k^{th} unit selected from sampling unit
3. Stratified Sampling - population into groups called strata
4. Cluster Sampling (heterogeneous groups)
 $\rightarrow$ few clusters chosen randomly
precision being less $\rightarrow$ based on availability
5. Judgement & Quota Sampling
 $\downarrow$
based on expertise
6. Snow ball sampling :- existing subjects recruit future subjects

7. Error
- $\rightarrow$ Sampling - diff b/w theoretical & Practical
 - $\rightarrow$ Non Sampling - manual errors

Estimators

$\rightarrow$ population

Estimator :- fixed but unknown numerical value

$$\hat{\theta} = T(x_1, x_2, \dots, x_n)$$

$T \rightarrow$ statistic

$$\hat{\mu} = \bar{x}$$

$$\sigma^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Estimator of population proportion

$$\hat{p} = \frac{x}{n} \text{ where } x \text{ is the number of subjects.}$$

Unbiasedness $E(\hat{\theta}) = \theta \longrightarrow$ true population parameter
 $\downarrow$
 expected value

If $\bar{x} = \mu \implies$ unbiased.

* COMPARING TWO ESTIMATORS

$$\hat{\mu}_1 = \frac{x_1 + x_2 + x_3}{3} \quad \hat{\mu}_2 = \frac{x_1 + 2x_2}{3}$$

Unbiased $\implies E[\hat{\mu}_1] = E[\hat{\mu}_2] = \mu$

variances $\text{Var}(\hat{\mu}_1) = \frac{\sigma^2}{3} \quad \text{Var}(\hat{\mu}_2) = \frac{5\sigma^2}{3}$

Hence $\hat{\mu}_1$ is more efficient than $\hat{\mu}_2$.

* Hypothesis Testing :

statistical hypothesis $\rightarrow$ assumption on population parameter.

Hypothesis testing decides whether sample data is supported or contradicted.

$\rightarrow$ 2 hypothesis — NULL and alternative $\rightarrow$ (no diff/effect) $\rightarrow$ (diff exists)

$\Rightarrow$ significance level 0.05 or 0.01

Errors $\rightarrow$ Type 1 error $\rightarrow$ rejects Null hypothesis which is True
 Type 2 error $\rightarrow$ accepting Null hypothesis when it is False.

{ power of the Test = $1 - \beta$ }

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Test (Z test)

Qn.) $N = 50$

$\bar{X} = 0.36$

$\sigma = 0.10$

$\mu_0 = 0.4$

$$Z = \frac{0.36 - 0.4}{0.10 / \sqrt{50}} \Rightarrow \frac{-0.04}{0.10 / 5\sqrt{2}}$$

$$\Rightarrow \frac{\frac{4}{100}}{\frac{1}{10} / 5\sqrt{2}} \Rightarrow \frac{4}{50\sqrt{2}}$$

$$\frac{-4\sqrt{2}}{50\sqrt{2}} < =$$

Z test (when sample size is more > 30) Soln $\Rightarrow -2.83$ & variance is known

T test $\rightarrow$ when variance is unknown

Chi square Test $\rightarrow$ observed & expected frequencies given

24/10/2025

MATH.* Statistic:-

Significance level (α) — probability of rejecting the Null hypothesis (H_0) when it is actually True.
 — It represents a risk of making Type I error.

Power of a test

↓ probability that the test rejects the Null Hypothesis.

Two-tailed Testtwo tailed $\mu = \mu_0$.

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0$$

one tailed when any one of the tests are accepted eg: $\mu > \mu_0$
 $\mu < \mu_0$.

* Confidence Interval — considers only mean/proportion.

One sample Z-Test:-

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

 $\bar{X}$ → mean μ_0 → population mean σ → std. deviation n → total no. of population or (count)

$H_1: \mu \neq 100$ (depicts one tailed)

Rejection of null hypothesis — when t is lesser than the calculated value

Left tailed $Z_{cal} < Z_{tab} \rightarrow \text{Reject } H_0.$

If 2 tailed consider $\Rightarrow |Z_{cal}|.$

Two sample Z Test:

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

σ_1, σ_2 , is the population standard deviation

Hypothesis is rejected when, t is lesser than the calculated value.

Two tailed = $\pm$ value

left tailed (single tailed) = $+$ value.

right tailed (single tailed) = $-$ value.

Problem 1:

A company manufactures resistors with a claimed mean resistance of 100Ω

Population std deviation $\sigma = 2.5 \Omega$

Random sample, $n = 60$, Sample mean $\bar{X} = 99.4 \Omega$

Soln:- Given

$$\sigma = 2.5 \Omega$$

$$n = 60$$

$$\alpha = 0.01$$

$$\bar{X} = 99.4 \Omega$$

$$\mu_0 = 100 \Omega$$

(ii) Check if True mean differs from 100Ω .

$$Z_{\text{test}} = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

$$\Rightarrow \frac{99.4 - 100}{2.5 / \sqrt{60}} \Rightarrow \frac{0.6}{2.5 / \sqrt{6 \times 10}}$$

$$\Rightarrow \frac{0.6}{2.5 / 2 \times \sqrt{15}}$$

$$\Rightarrow \left(\frac{0.6 \times 2 \times \sqrt{15}}{2.5} \right)$$

$$Z_{\text{cal}} \Rightarrow -1.86 \quad (\text{Calculated value of } Z_{\text{test}})$$

 $Z_{\text{tabulated}}$ for

$$2 \text{ tails} \Rightarrow \alpha/2 \quad \alpha \rightarrow 0.01$$

$$\text{then } \alpha/2 = 0.005$$

 α determines whether single or 2 tailed

$$Z_{\text{tab}} = 2.58$$

$$|z_{cal}| < z_{tab}$$

i.e. $1.86 < 2.58$.

Hence the hypothesis is failed to be rejected

i.e. the hypothesis is accepted

Hence, the NULL Hypothesis is accepted.

Hence True mean differs from 100.2

(P^o)

$$Z\text{-test} \Rightarrow \bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$z_{test} = 99.4 + \frac{2.58 \times 2.5}{\sqrt{60}}$$

$$\text{When } 99.4 + \frac{2.58 \times 2.5}{\sqrt{60}} = 98.57$$

$$99.4 - \frac{2.58 \times 2.5}{\sqrt{60}} = 100.23$$

Hence the interval is $(98.57, 100.23)$

Soln:- Mean lies in the above region, i.e.

$(98.57, 100.23)$, hence it cannot be rejected.

applied + and - values separately.

applicable for 2 tail and 2 right tail

$$\left\{ \begin{array}{l} |z_{cal}| < |z_{tab}| \Rightarrow \text{Null hypothesis is accepted} \\ |z_{cal}| < |z_{tab}| \end{array} \right.$$

Right tail & 2 tail
 Left tail $\rightarrow$ -ve.

(iii) Interpret ~~of~~ result
mean $\rightarrow$ 100%

$$\mu_0 = 250 \text{ ml}$$

$$n = 36 \text{ cups.}$$

$$\sigma = 5.$$

$$\bar{X} = 247 \text{ ml}$$

$$H_0 \Rightarrow \mu = 250 \text{ ml}$$

$$\mu \neq 250 \text{ ml.}$$

$$\begin{array}{r} 3.6 \\ 5 \overline{) 18} \\ \underline{-15} \\ 30 \\ \underline{30} \end{array}$$

$$Z = \frac{247 - 250}{5/\sqrt{36}} \Rightarrow \frac{-3}{5/6} \Rightarrow$$

$$\frac{-3 \times 6}{5} = \frac{-18}{5} = \underline{\underline{-3.6}}$$

$$Z_{\text{cal}} = -3.6.$$

$$Z_{\text{tab}} \text{ at } \alpha = 0.05$$

$$\Rightarrow \underline{\underline{2.40}}$$

$$Z_{\alpha/2} = 1.96.$$

Calculated value $>$ Tabulated value of
Z thus, the hypothesis is rejected.

(*) 95% confidence.

$$CI = \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$\Rightarrow 247 + 1.96 \times \frac{5}{6}$$

$$\Rightarrow (247.87, 248.63)$$

250 ml is not in the range of the interval $(247.87, 248.63)$ and hence the hypothesis proposed is rejected.

2 Sample z-test :-

Battery A and B

$$\sigma_A = 3.5 \text{ hours.}$$

$$\sigma_B = 4.0 \text{ hours.}$$

Q1 - says $\Rightarrow$ mean lifetime

Null Hypothesis : if ~~the~~ Mean lifetime of battery of brand A, μ_1 and Mean lifetime of battery of brand B, μ_2 .

$$CI = \bar{X} \pm z_{\alpha} (\sigma / \sqrt{n}).$$

$$z_{\text{calculated}} \quad z_{\text{test}} = \frac{41.8 - 39.6}{\sqrt{3.5^2 / 40 + 4^2 / 50}} \Rightarrow \underline{\underline{2.86}}$$

$$(z_{\text{tabulated}} \text{ with } \alpha \text{ value}). \quad z_{\alpha \text{ value}} = \frac{z_{0.01}}{2} = \underline{\underline{2.58}}$$

$z_{\text{cal}} > z_{\text{tab}}$, Hence the Null hypothesis, H_0 is rejected, Hence their battery lifetime differ.

P.T.O.

Z-test

2tailed :-

$$(\bar{x}_1 - \bar{x}_2) \pm Z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

for 2 tailed, we consider.

2 values of mean population
along with 2 different values of
standard deviation and n value,
count of the population.

$$\bar{x}_1 = 41.8 \quad \bar{x}_2 = 39.6 \quad Z_{\alpha/2} = 2.58$$

$$Z_{tab} = \sigma_1 = 3.5 \quad \sigma_2 = 4.0 \quad n_1 = 40 \quad n_2 = 50$$

$$CI \quad Z_{\alpha/2} = (41.8 - 39.6) \pm 1.96 \sqrt{\frac{(3.5)^2}{40} + \frac{(4.0)^2}{50}}$$

$$\Rightarrow 2.2 \pm 1.96 \times 0.79$$

$$\Rightarrow 2.2 + (1.96 \times 0.79)$$

Qn:- Ho-

or

$$2.2 - (1.96 \times 0.79)$$

$$\text{result} = (0.649, 3.751)$$

range of CI would have
be $\Rightarrow (0.649, 3.751)$

↓
result

$$\text{Approx } (0.65, 3.75)$$

Brand A lasts significantly in Brand B.

Q1) A simple random sampling survey in respect of monthly earnings of semi-skilled workers in two cities gives the following statistical information.

City	Mean Monthly earnings	SD of sample data of monthly earnings	Size of sample
A	695 Rs.	40	200
B	710 Rs.	60	175

Test the hypothesis at 5% significance level that there is no difference b/w monthly earnings of workers of 2 cities.

Soln

$$\bar{X}_A = 695$$

$$\bar{X}_B = 710$$

$$\sigma_A = 40$$

$$\sigma_B = 60$$

$$n_A = 200$$

$$n_B = 175$$

Hypothesis

$$H_0 = \mu_A = \mu_B$$

$$H_1 = \mu_A \neq \mu_B$$

(2-tailed)

$$\begin{array}{r} 610 \\ - 710 \\ \hline -695 \\ 15 \end{array}$$

$$Z = \frac{\bar{X}_A - \bar{X}_B}{\sqrt{\frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}}} = \frac{695 - 710}{\sqrt{\frac{40^2}{200} + \frac{60^2}{175}}}$$

$$\Rightarrow \frac{-15}{\sqrt{\frac{1600}{200} + \frac{3600}{175}}}$$

$$= \frac{-15}{\sqrt{8 + 20.57}} = \frac{-15}{\sqrt{28.57}} = \frac{-15}{5.34} = -2.809$$

$$\frac{Z_{\alpha}}{2} = \frac{2.05}{2} = 1.96$$

$$|Z_{cal}| > Z_{tab}$$

Hence reject H_0 .

$$|Z_{cal}| > Z_{tab} \rightarrow \text{reject } H_0.$$

* Therefore, there is a difference b/w monthly earnings of 2 cities. (Since Null hypothesis is rejected)

Qn).

Null hypothesis $\Rightarrow$ at least 1200, ≥ 1200

Alternate hypothesis $\Rightarrow < 1200$.

$$n = 40$$

$$\bar{X} = 1185$$

$$1200$$

$$\sigma = 30$$

$$\mu_0 = 1200$$

$$-1185$$

$$15$$

$H_0 \geq 1200 \rightarrow$ Null hypothesis

$H_1 < 1200 \rightarrow$ Alternate hypothesis.

$$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{1185 - 1200}{\frac{30}{\sqrt{40}}}$$

$$\Rightarrow \frac{-15}{\frac{30}{\sqrt{40}}} = -\frac{15 \times \sqrt{40}}{30}$$

$$Z_{test} = -\sqrt{10}$$

$$Z_{tab} = Z_{\alpha} = Z_{0.05} = \underline{-1.645}$$

$$|Z_{cal}| = 3.65 > 1.645 = |Z_{tab}|$$

Hence, H_0 is rejected.

Hence, Mean life is less than 1200.

③. Determine whether factory's claim is supported by the sample.

Q

T-test

- Used to compare sample mean.

Types of t-test

- One sample t-test
- Two sample t-test

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0$$

* One sample t-test

$$t = \frac{\bar{X} - \mu_0}{s / \sqrt{n}}$$

Compare $|t|$ to t_{n-1} and $\alpha/2$ by T-table

In one sample degree = t-test

$$t_{n-1} \alpha/2$$

* 2 sample t-tests.

↳ Check if population variances are equal or not.

* If variances are equal:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Pooled standard deviation :-

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Degrees of freedom = $df = n_1 + n_2 - 2$

* Unequal variances

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Degrees of freedom :- Welch-Satterthwaite approximation

$$df \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{\left(\frac{s_1^2}{n_1} \right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2} \right)^2}{n_2 - 1}}$$

One tail & 2 tail considered in t-test.

Qn) A process scheduler claims mean CPU utilization = 70%. Sample of 10 users: 68, 72, 69, 70, 65, 71, 68, 74, 69, 67. At $\alpha = 0.05$, verify claim.

Soln :- Null hypothesis H_0 :- Mean CPU utilization = 70%

Alternate hypothesis H_1 :- Mean CPU utilization $\neq 70\%$

$$\bar{X} = \frac{68 + 72 + 69 + 70 + 65 + 71 + 68 + 74 + 69 + 67}{10}$$

$$\bar{X} = 69.3$$

$$\text{Standard deviation} = \sqrt{\frac{\sum (X - \bar{X})^2}{n-1}}$$

$$\begin{array}{r} 22 \overline{) 47} \\ \underline{44} \\ 30 \\ \underline{22} \\ 80 \\ \underline{74} \\ 60 \\ \underline{56} \\ 40 \\ \underline{38} \\ 20 \end{array}$$

$$\begin{aligned} & (68 - 69.3)^2 + (72 - 69.3)^2 + (69 - 69.3)^2 + \\ & (70 - 69.3)^2 + (65 - 69.3)^2 + (71 - 69.3)^2 + \\ & (68 - 69.3)^2 + (74 - 69.3)^2 + (69 - 69.3)^2 \\ & + (67 - 69.3)^2 \end{aligned}$$

9.

$$\begin{array}{r} 26.03 \\ 3 \overline{) 60.1} \\ \underline{6} \\ 0010 \\ \underline{9} \\ 10 \end{array}$$

$$- (32 + 23 + 0.3 + 0.7 + 4.3 + 4.7 + 3)$$

$$\begin{aligned} & \sum 1.69 + 7.29 + 0.09 + 0.49 + 18.49 + 289 + \\ & 1.69 + 22.09 + 0.09 + 5.29. \end{aligned}$$

9

$$\Rightarrow \sqrt{\frac{1}{9} \times (60.1)} \Rightarrow \underline{\underline{2.58}}$$

$$\Rightarrow \text{soln} = \underline{\underline{2.58}}$$

$$\text{thus } T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} = \frac{69.3 - 70}{2.58/\sqrt{10}} = 0.857$$

$$t_{cal} =$$

$$\text{degree of freedom is } n-1 = 10-1 = 9$$

$$\Rightarrow 10-1 = 9$$

$$\text{just for one tail}$$

$$= 1.8$$

$$t_{tabulated} = t_{9, \frac{0.05}{2}} = t_{9, 0.025} \rightarrow \text{one tail problem}$$

Hence $t_{cal} = -0.858$.

$t_{tab} = 2.262$.

$$(|-0.858|) < t_{tab} (2.262)$$

Hence failed to Reject H_0 .

Hence H_0 is accepted. Mean CPU Life
time is 70% (accepted)

MATHST-test continuationIndependent Samples t-test: (Model Accuracy)

A researcher evaluates 2 ML models using 10-fold cross-validation:

Model A: 91 89 90 92 88 93 90 91 89 90

Model B: 88 85 87 86 88 89 86 87 85 86

Test whether there is significant difference in mean accuracies at $\alpha = 0.05$. Assume the two samples are independent & variances are equal.

	X_A	$X_A - \bar{X}_A$	X_B	$X_B - \bar{X}_B$	$(X_A - \bar{X}_A)^2$	$(X_B - \bar{X}_B)^2$
1	91	0.7	88	1.3	0.49	1.69
2	89	-1.3	85	-1.7	1.69	2.89
3	90	-0.3	87	0.3	0.09	0.09
4	92	1.7	86	-0.7	2.89	0.49
5	88	-2.3	88	1.3	5.29	1.69
6	93	2.7	89	2.3	7.29	5.29
7	90	-0.3	86	-0.7	0.09	0.49
8	91	0.7	87	0.3	0.49	0.09
9	89	-1.3	85	-1.7	1.69	2.89
10	90	-0.3	86	-0.7	0.09	0.49
					<u>20.10</u>	<u>16.1</u>

$$\bar{X}_A = 90.3$$

$$\bar{X}_B = 86.7$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{sp \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\begin{array}{r} 4.528 \\ 8 \overline{) 36.2007} \\ \underline{-32} \\ 42 \\ \underline{-40} \\ 20 \\ \underline{-16} \\ 40 \\ \underline{-40} \\ 07 \end{array}$$

$$S_A^2 = \frac{\sum (X_A - \bar{X}_A)^2}{n_A - 1} \Rightarrow \frac{20 \cdot 10}{9} = 2.233 \quad \begin{array}{l} n_A = 10 \\ n_A - 1 = 9 \end{array}$$

$$S_B^2 = \frac{\sum (X_B - \bar{X}_B)^2}{n_B - 1} \Rightarrow \frac{16 \cdot 1}{9} \Rightarrow 1.789$$

Thus $S_A^2 = 2.2333$ $S_B^2 = 1.789$

Qn)

$$t = \frac{\bar{X}_A - \bar{X}_B}{sp \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \Rightarrow \frac{90.3 - 86.7}{sp \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Qn)

$$sp = \sqrt{\frac{(n_A - 1) S_A^2 + (n_B - 1) S_B^2}{n_A + n_B - 2}}$$

$$\begin{array}{r} 63 \\ 18 \\ 2.88 \\ 1.789 \\ \times 9 \\ \hline 16.101 \\ \times 2 \\ \hline 32.202 \\ \times 9 \\ \hline 2.122 \\ 2.2333 \\ \times 9 \\ \hline 200799 \end{array}$$

$$\begin{array}{r} 20.0997 \\ 16.1010 \\ \hline 36.2007 \end{array}$$

$$\Rightarrow \sqrt{\frac{(9 \times 2.2333) + (9 \times 1.789)}{18}} \Rightarrow \sqrt{\frac{20.0997 + 16.101}{18}} \Rightarrow \sqrt{\frac{36.2007}{18}} \Rightarrow \sqrt{2.01115}$$

$$t = \frac{90.3 - 86.7}{\sqrt{14.528} \sqrt{\frac{1}{10} + \frac{1}{10}}} \Rightarrow \underline{\underline{1.417}}$$

$$\text{degree of freedom } df = n_A + n_B - 2 \\ \Rightarrow 10 + 10 - 2 \\ = 18.$$

$$t_{18, 0.05} = 2.101.$$

$$t_{cal} = 5.689$$

$$t_{cal} = 5.689 > t_{tab} = 2.101.$$

Hence $\Rightarrow$ there is significance diff.
i.e. the Hypothesis H_0 is rejected.

Qn-2 Equal variances {Pool t test}

Qn-2 A group of seven chickens reared on a high protein diet of 12, 15, 11, 16, 14, 4 and 16 ounces; a second group of five chickens of same age similarly treated except that they receive a low protein diet of 8, 10, 14, 10 and 13 ounces. Test at 5 per cent level whether there is significant evidence that additional protein has increased the weight of the chickens.

According to the above question - there are 2 hypothesis

Hypothesis

$H_0 \rightarrow$ weights of high protein and low protein diet chickens are same.

$H_1 \rightarrow$ alternate hypothesis: Weight of high protein diet chickens are weight of low protein diet chickens.

$$\text{degree of freedom} = n_1 + n_2 - 2$$

$$\Rightarrow 7 + 5 - 2 = 12 - 2 = 10.$$

5% Significance level.

$$t_{10, 0.05} \Rightarrow \underline{\underline{t_{0.05, 10} = 1.812}} \quad (t_{\text{tabulated}})$$

$$\text{Similarly} \Rightarrow t_{\text{calculated}} \Rightarrow \frac{\bar{X}_1 - \bar{X}_2}{S_1^2}$$

$$\frac{S_1^2}{6} = \frac{22}{6} = 3.67.$$

X_1	$x_1 - \bar{X}_1$	$(x_1 - \bar{X}_1)^2$	X_2	$(x_2 - \bar{X}_2)$	$(x_2 - \bar{X}_2)^2$
12	-2	4	8	-3	9
15	1	1	10	-1	1
11	-3	9	14	3	9
16	2	4	10	-1	1
14	0	0	13	2	4
14	0	0			
16	2	4			
					<u>24</u>
$\bar{X}_1 = 14$		<u>22</u>	$\bar{X}_2 = 11$		

$$S_2^1 = \frac{22}{7} = 3.67$$

$$S_2^2 = \frac{24}{4} = 6$$

$$\text{Hence, } S_p^2 = \frac{6 \cdot (3.67) + 4(6)}{10}$$

$$S_p^2 = 4.6$$

$$S_p = \sqrt{4.6}$$

$$t = \frac{14 - 11}{\sqrt{4.6} \sqrt{\frac{1}{7} + \frac{1}{7}}} \Rightarrow \underline{\underline{2.389}}$$

$$\text{Hence } t_{cal} \ 2.389 > t_{tab} = 1.812$$

Hence, H₀ is rejected, i.e. H₁ → true.

Thus Weight of high protein diet chickens are more than weight of low protein diet chickens.